

Krystal: Making Web3 accessible to all with simplified digital wallets

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ABOUT KRystal

With Google Cloud is accelerating development of new features that enable businesses to take advantage of emerging trends, centered around simple digital wallets.



About Krystal

Founded in April 2021, Krystal provides a platform that enables users to access a range of Web3 services through the web and a mobile application which supports 10+ blockchain networks. The business has offices in Singapore and Vietnam and has to date served more than 1.8 million users worldwide.

**Google
Cloud
results**

Enab
incor

Industries: Financial Services

- Reduced development time by 50% with microservices architecture on GKE
- Enables the business to focus on product development and evaluating new trends rather than infrastructure administration
- Supported wide fluctuations in users and query numbers from around the globe with no disruption
- Enabled seamless growth to a maximum of 1 million monthly active users

new
trend
prod
servi

Location: Singapore

Google Cloud (<https://cloud.google.com/>)

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BigQuery (<https://cloud.google.com/bigquery/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

For Sunny Jain and Jarvis Nguyen, a simple digital wallet presented a compelling opportunity to realize the potential of Web3 for consumers. Jain, a five-year veteran of the cryptocurrency market with stints at businesses such as crypto trading and liquidity hub Kyber Network and Nguyen, who had delivered a decentralized game on the Ethereum blockchain, used their contacts and experience to start Krystal (<https://krystal.app/>), a digital wallet that supports Web3 services across multiple blockchains.

The Krystal digital wallet enables users to manage their portfolio, trade assets, earn passive income and participate in cryptocurrency launches with crowdfunding through its IDO platform, KrystalGO.

Jain and Nguyen have grown Krystal from 12 to 50 people in less than two years. Its 40-strong technology team is based in Vietnam while compliance, legal and finance teams operate from headquarters in Singapore.

The business is running Krystal on Google Cloud (<https://cloud.google.com/>) to execute a five-year business plan based on growth in Web3, a version of the web that passes control over access and ownership of sections of the web from corporates to users, does not require permission from authorities to access services, and, by enabling direct virtual transactions between two parties, eliminates the need to trust intermediaries.

By simplifying the digital wallets that act as entry points to Web3 initiatives such as blockchain and cryptocurrency, Krystal can power its own growth by

serving customers' Web3 needs in an easy, seamless manner. "When you use our wallet, we don't expect you to know anything about cryptocurrency or the blockchain—we aim to mirror the experience of running accounts on social media or e-commerce services," explains Jain. "We aim to provide the simplest wallet in the market."

Improving scalability, reliability and performance

The business had started operations with another cloud provider, but soon realized it needed more scalability, reliability and higher performance to support forecast growth. The Krystal leadership team also wanted to break down its monolithic application architecture into a microservices architecture that would allow teams to add new features and update its products and services faster.

After evaluating services such as [BigQuery](https://cloud.google.com/bigquery) (<https://cloud.google.com/bigquery>) to support its data architecture and [Google Kubernetes Engine](https://cloud.google.com/kubernetes-engine) (<https://cloud.google.com/kubernetes-engine>) to provide a managed, scalable containerized environment in which to run its services, Krystal opted for Google Cloud and migrated all its workloads seamlessly to the new platform in March 2022.

"We were able to move seamlessly across to Google Cloud due to its simplicity and the skilled support and advice from the Google Cloud account team," says Nguyen.

BigQuery enables Krystal to capture data from the blockchain and run queries quickly, to help users make important decisions based on time-sensitive information such as token pricing and user balances. "With Google Cloud products such as BigQuery and Google Kubernetes Engine in our arsenal, we could build a Web3 digital wallet platform and application—a project we didn't think was possible just a few years ago," says Jain.

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Executive Officer, Krystal

The ability to pivot quickly in line with Web3 changes

By running a microservices architecture in Google Kubernetes Engine, Krystal has increased the speed of development by at least 50%. "Each team can proceed with its own tasks without relying on another team to deploy as is the case in a monolithic architecture," explains Nguyen.

With the flexibility of Google Cloud, Krystal can also pivot rapidly to address ongoing changes in the Web3 environment. "Every two to three months, everything changes and each new trend requires different responses from a technology standpoint," says Jain. "NFTs may be the hot trend, then liquidity as a chain, then derivatives in the Web3 space." The managed services and ease of use of Google Cloud means developers can focus on enabling each trend for its customers.

Incorporating new trends on an agile platform

Krystal's leadership team can now devote its energy to determining which trends may be around in the longer-term, and on positioning the business to keep pace with innovation and become more forward-looking.

"We've not had any significant downtime with Google Cloud, even as we grew our employee numbers fourfold and surged to a maximum of one million monthly active users," says Jain. "We've also been able to align our costs with growth, which is a considerable benefit for a fledgling business like ours."

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"Along with our easy to use digital wallet, our technology infrastructure is a point of differentiation for us. The speed and scalability we achieve with Google Cloud are impeccable. We don't worry about scaling, we just get on with building things."

Krystal now seamlessly accommodates widely fluctuating numbers of users from around the world, including Africa and Southeast Asia, and the associated changes in query numbers, without disruption.

In addition, the ease of use of Google Cloud and the clear documentation that comes with its products has made it easy for engineers to pick up and use them, with minimal training required. "Any engineer can start using Google Cloud after one or two days of training and investigation," says Jain. "We also have very strong support from the Google Cloud team to ensure our infrastructure and data architecture meets our needs."

With Google Cloud, Krystal is well positioned for further growth and expansion. "Because we want to focus our limited resources on building products

and services, we need a provider that can deliver a stable, scalable, robust infrastructure and managed services," says Jain. "Google Cloud is the perfect organization to help us realize our ambitions."

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